CHAPTER 3: METHODS

Senior Honors Thesis — Department of Environmental Science
"Microplastic Abundance and Polymer Composition Along an Urban-to-Rural Gradient of the Tamsen River"
Author: Alina Brooks
Advisor: Dr. Raymond Otieno
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3.1 Overview of Study Design

This chapter describes the procedures used to collect, process, and analyze surface water and sediment samples from the Tamsen River between June and September. The study was designed to test the hypothesis that microplastic concentration increases with proximity to dense urban land use and to wastewater treatment plant (WWTP) outfalls, and that the dominant particle shapes and polymer types shift along that gradient. The design combines field sampling at fixed stations, laboratory separation of particles from organic and mineral material, visual sorting under a stereomicroscope, and polymer identification of a subset of particles using Fourier-transform infrared spectroscopy (FTIR).

Because microplastic research is highly vulnerable to contamination — synthetic fibers from clothing, laboratory equipment, and indoor air can easily outnumber the particles actually present in a sample — a substantial portion of this chapter is devoted to quality assurance and quality control (QA/QC). The procedures described here were developed in part from published protocols (Masura et al. 2015; Hidalgo-Ruz et al. 2012) and in part through trial and error during a pilot season the previous autumn, as described in Section 3.6.

3.2 Study Area and Site Selection

The Tamsen River drains approximately 1,140 square kilometers, flowing roughly southeast from forested headwaters through mixed agricultural land and then through the city of Port Hallam (population approximately 210,000) before entering the estuary. Land use within the watershed was classified using the most recent regional land cover dataset, and sampling stations were selected to represent a gradient from minimally disturbed to heavily urbanized conditions.

Eight stations were established:

- T1 (Ashby Falls): forested headwaters, above all known point sources.
- T2 (Millbrook Road bridge): mixed forest and pasture.
- T3 (Carver Farm access): agricultural, downstream of two dairy operations.
- T4 (Lindell Park): suburban edge of Port Hallam, upstream of the WWTP.
- T5 (WWTP outfall, 50 m downstream): directly influenced by treated effluent.
- T6 (Harbor Street): urban core, adjacent to stormwater drains.
- T7 (Old Rail Yard): urban, downstream of an industrial area.
- T8 (Estuary mouth): tidal influence, integrating the whole watershed.

Stations were chosen to be accessible on foot and to allow sampling from a bridge or stable bank. Station T5 was positioned 50 meters downstream of the outfall after consultation with the plant operator, who confirmed that effluent was well mixed by that distance under normal flow conditions. Coordinates for each station are given in Appendix A.

3.3 Field Sampling

3.3.1 Surface Water

Surface water was sampled on six occasions at each station, approximately every three weeks, for a total of 48 surface water samples. At each visit, 100 liters of surface water were collected using a stainless steel bucket and poured throug

h a stack of stainless steel sieves. Originally the plan was to use a manta traw l, which is standard in marine studies, but the river was too shallow at most st ations during summer low flow, so the bucket-and-sieve method was used throughou t to keep sampling consistent.

The sieve stack consisted of a 5 mm sieve on top (to remove large debris and def ine the upper size limit for microplastics) and a 300 µm sieve below. The materi al retained on the 300 µm sieve was rinsed into pre-cleaned glass jars with filt ered deionized water, capped with aluminum foil under the lid, and stored on ice until returning to the lab. The decision to use 300 µm rather than a finer mesh was a compromise: finer meshes clogged within a few liters at the turbid downst ream stations, and 300 µm allows comparison with the majority of published fresh water surveys. This means that the smallest particles, which may be the most num erous, are not captured, and the results should be interpreted as a conservative estimate.

Water temperature, dissolved oxygen, conductivity, and turbidity were recorded a t each visit using a handheld multiparameter probe. Flow conditions were noted q ualitatively (low, moderate, high) and later matched to daily discharge from the nearest gauging station.

3.3.2 Sediment

Sediment was collected once per station in mid-August, at three points across th e channel, using a stainless steel scoop to a depth of approximately 5 cm. Rough ly 500 g wet weight was collected at each point and the three subsamples were co mposited. Sediment samples were stored in glass jars at 4 °C until processing.

3.4 Laboratory Processing

3.4.1 Wet Peroxide Oxidation

Surface water samples contained large amounts of organic material such as algae, leaf fragments, and insect parts, which would otherwise be confused with plasti c during visual sorting. Organic matter was removed by wet peroxide oxidation (W PO). Each sample was dried at 60 °C, then 20 mL of 0.05 M Fe(II) solution and 20 mL of 30% hydrogen peroxide were added. The mixture was heated to 75 °C on a ho tplate and additional peroxide was added in 20 mL increments until no visible or ganic material remained. The temperature was kept below 75 °C because higher tem peratures can deform some polymers, particularly polyethylene and polypropylene.

3.4.2 Density Separation

After digestion, plastics were separated from mineral material by density separa tion. Sodium chloride is commonly used but its density (about 1.2 g/cm³) is too low to float denser plastics such as PET and PVC. A zinc chloride solution adjus ted to 1.6 g/cm³ was therefore used. Samples were stirred, allowed to settle for 24 hours in a glass separation funnel, and the supernatant was filtered onto 0. 7 µm glass fiber filters. The zinc chloride solution was recovered by filtration and reused up to five times; its density was checked with a hydrometer before e ach use.

Sediment samples were processed in the same way after drying and sieving, except that density separation was done twice for each sample to improve recovery.

3.4.3 Visual Sorting

Filters were examined under a stereomicroscope at 10-40x magnification. Particle s were classified by shape into five categories: fibers, fragments, films, foams , and pellets/beads. Color and approximate length were also recorded. Criteria f or identifying a particle as likely plastic followed Norén (2007): no visible ce llular or organic structures, uniform thickness for fibers, and clear homogeneou

s color. A hot needle test was used on ambiguous particles — plastic particles m elt or curl when touched with a heated needle, while organic particles do not.

3.5 Polymer Identification

A random subsample of 20% of particles from each sample (minimum 10 particles wh ere available) was analyzed by attenuated total reflectance FTIR on the departme nt's spectrometer. Spectra were compared against a reference library, and a matc h quality above 70% was accepted as a positive identification. Particles below r oughly 500 µm were too small for reliable ATR measurement with the available equ ipment, which means polymer data are biased toward larger particles. This limita tion is discussed further in Chapter 5.

3.6 Quality Assurance and Contamination Control

Contamination control was the single most time-consuming aspect of this project. During the pilot season, procedural blanks — filters run through every step of processing without any sample — contained an average of 14 fibers each, which wa s higher than the counts in several real samples from the headwater stations. Mo st of these blank fibers were blue and black polyester, and after some investiga tion the main source turned out to be the synthetic fleece and lab coats worn in the lab. The lab's standard-issue coats were a poly-cotton blend.

For the main season, the following measures were adopted:

1. All personnel wore 100% cotton lab coats (purchased specifically for this pro ject) and avoided synthetic clothing on processing days. Cotton fibers, if they appeared, could be identified and excluded by FTIR.
2. All work with open samples was done inside a laminar flow hood that had been wiped down with filtered ethanol.
3. Glassware was rinsed three times with filtered deionized water and covered wi th aluminum foil immediately.
4. Plastic equipment was eliminated wherever possible; where unavoidable (e.g., wash bottles), the polymer was recorded so that any matching particles could be flagged.
5. A procedural blank was run with every batch of six samples, and an open petri dish with a damp filter was left beside the microscope during every sorting ses sion to capture airborne fibers.

With these measures in place, procedural blanks averaged 1.8 fibers per filter, and airborne controls averaged 0.6 fibers per session. Final sample counts were blank-corrected by subtracting the mean blank count of the corresponding batch, by shape and color category.

Recovery efficiency was estimated by spiking three clean water samples with 50 k nown particles each (green polyethylene fragments and red PET fibers cut to 1-2 mm). Mean recovery was 87% for fragments and 79% for fibers. Counts were not adj usted for recovery, but these figures are reported to help interpret the results .

3.7 Data Analysis

Microplastic concentrations in surface water are reported as particles per cubic meter, and in sediment as particles per kilogram dry weight. Differences among stations were tested using a Kruskal-Wallis test, since count data were strongly right-skewed and did not meet the assumptions of ANOVA even after log transform ation. Pairwise comparisons used Dunn's test with Bonferroni correction.

To test the relationship between land use and concentration, the percentage of u rban land within the upstream catchment of each station was calculated in GIS an d used as a predictor in a generalized linear mixed model with a negative binomi al distribution, with sampling date as a random effect. Discharge was included a

s a covariate because higher flows may either dilute particles or resuspend them from sediment.

All analyses were performed in R (version 4.3) using the packages lme4, glmmTMB, and FSA. Code and raw data are archived in the department repository (see Appendix C).

3.8 Limitations

Several limitations of the methods should be acknowledged. First, the 300 µm lower size limit excludes small microplastics, which are likely abundant downstream of the WWTP. Second, FTIR was limited to particles larger than about 500 µm, so polymer composition may not represent the full particle population. Third, sediment was sampled only once, so temporal variation in sediment is unknown. Finally, because surface water was sampled with a bucket from the bank or bridge, samples may not represent the full water column, and buoyant particles may be overrepresented relative to denser ones.

Despite these limitations, the consistent methods across all eight stations allow meaningful comparison along the gradient, which is the central aim of this thesis.

[Note to self: Dr. O wants Table 3.1 with all station coordinates + land use %, and a flowchart figure for lab processing. Also cite the zinc chloride disposal protocol from EHS.]